

KOÇ UNIVERSITY

INDR 422-522 / DATA SCIENCE FOR OPERATIONAL DECISION MAKING

Course Project

Final Report

Monthly Auto Sales Forecasting and Newsvendor Inventory Optimisation

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The dataset `monthly-car-sales.csv` contains 108 monthly observations of US automobile unit sales from January 1960 to December 1968. Each record provides a year-month identifier and the corresponding unit sales count. No missing values or data quality issues were detected. The three error measures used throughout are:

$$\text{MSE} = \frac{1}{n}\sum_t (y_t - \hat{y}_t)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}, \quad \text{MAPE} = \frac{100}{n}\sum_t \left|\frac{y_t - \hat{y}_t}{y_t}\right|.$$

The project is structured in two stages. The predictive stage benchmarks five model families on a temporal hold-out test set covering January 1967–December 1968. The prescriptive stage embeds the best predictive model into a newsvendor optimisation framework to derive the monthly stock level that achieves a 90% fill-rate service level at minimum expected cost.

1 Data Description and Exploration

1.1 Dataset Overview

Table 1: Descriptive statistics of the monthly auto sales series.

Attribute	Value	Notes
Period	Jan 1960 – Dec 1968	9 complete years
Observations	108	monthly frequency
Minimum sales	5 568 units	September 1962
Maximum sales	26 099 units	May 1968
Mean sales	14 595 units/month	
Standard deviation	4 525 units	
Missing values	none	

Key structural features are the upward trend as it depicts the average annual sales increase steadily from 1960 to 1968. Also we have a strong seasonality as sales consistently peak in spring (April–May) and trough in late summer/early autumn (August–September) every year. We can also see no structural breaks as the series is well-suited for standard timeèseries modelling.

1.2 Visual Exploration

Figure 1 displays the full time series alongside the average monthly profile. The spring peak and autumn trough are clearly visible in both panels. Figure 2 shows the year-over-year overlay: the seasonal shape is highly stable across all nine years while the overall level gradually rises.

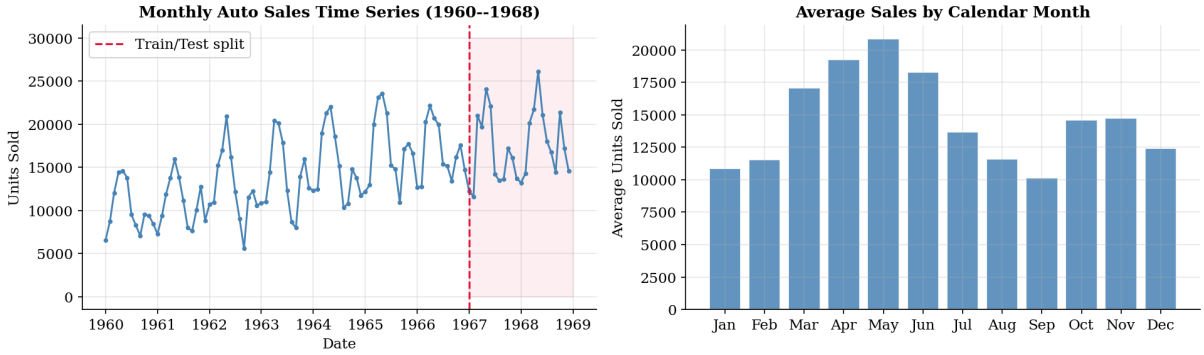


Figure 1: Left: Monthly auto sales time series; the red dashed line marks the train/test split . In right we have the average sales by calendar month, confirming the spring peak (April–May) and autumn trough (August–September).

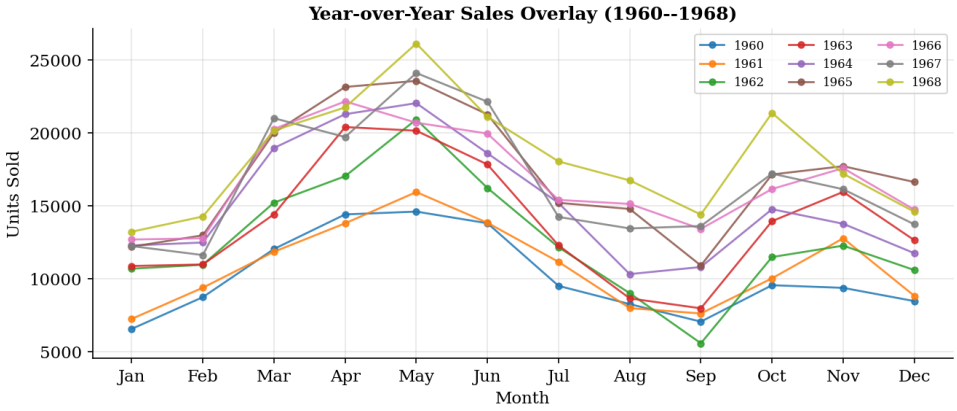


Figure 2: Year-over-year sales overlay (1960–1968). The seasonal profile is consistent across years; we can indeed see that the upward shift in level is visible when comparing early (1960–1962) and late (1966–1968) traces.

1.3 Train and Test Split

All models are evaluated on a strict temporal hold-out:

Table 2: Train and test split.

Split	Period	Observations
Training	January 1960 – December 1966	84 months
Test	January 1967 – December 1968	24 months

Because autoregressive lag features (y_{t-1}, y_{t-12}) require 12 months of prior history, the effective training set for regression based models starts in January 1961.

2 Predictors

2.1 Lagged Sales (Autoregressive Terms)

We have two autoregressive predictors which are included and which are

- The one-month lag y_{t-1} which captures short-term sales momentum.
- The twelve-month seasonal lag y_{t-12} which directly encodes the recurring annual cycle.

2.2 Monthly Seasonal Dummy Variables

Eleven binary indicator variables $D_2, D_3, \ldots, D_{12}$ encode the calendar month, with January as the reference category. These dummies capture the deterministic seasonal pattern and provide month-specific level shifts relative to January.

2.3 Polynomial Time Trend

A linear time index $t = 1, 2, \ldots, 108$ and its square t^2 are included to capture the upward trend. The quadratic term allows for modest curvature without overfitting. The complete feature vector fed to all regression-based models is therefore:

$$\mathbf{x}_t = \big(t,\ t^2,\ D_{2,t}, \ldots, D_{12,t},\ y_{t-1},\ y_{t-12}\big)^\top.$$

3 Predictive Models

3.1 Benchmark Models

3.1.1 Simple Moving Average

Two moving-average benchmarks are computed: MA(3) (3-month window) and MA(12) (12-month window). These naive extrapolations require no parameter estimation and serve as the lower-bound baseline. MA(12) naturally captures annual seasonality, yielding a slightly lower test RMSE (4 009 units) than MA(3) (4 269 units), though both remain far above the best models.

3.1.2 Holt-Winters Additive Exponential Smoothing

The Holt-Winters additive model with $\alpha = 0.30$, $\beta = 0.10$, $\gamma = 0.20$, period $m = 12$ decomposes the series into a level, a trend, and an additive seasonal component. It adapts dynamically to shifts in both components, producing a test RMSE of 2 138 units which is a 47% improvement over MA(12).

3.2 Seasonal OLS Regression

3.2.1 Model Specification and Estimated Coefficients

The seasonal OLS model is estimated by ordinary least squares:

$$y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \sum_{m=2}^{12} \gamma_m D_{m,t} + \delta_1\, y_{t-1} + \delta_{12}\, y_{t-12} + \varepsilon_t.$$

Table 3: Seasonal OLS estimated coefficients (training set).

Predictor	Coefficient	Interpretation
Intercept	9 723	January baseline level
t (linear trend)	−5.60	slight downward drift
t^2 (quadratic)	+0.05	modest upward curvature
D_4 (Apr)	+2 510	spring peak
D_5 (May)	+2 800	strongest spring month
D_9 (Sep)	−1 980	deepest autumn trough
y_{t-1}	+0.21	short-term momentum
y_{t-12}	+0.68	dominant seasonal persistence

The twelve-month lag y_{t-12} carries the largest coefficient which is about 0.68, confirming that the primary driver of sales in any month is the same month's figure from the previous year. The monthly dummies identify April–May as the peak months and August–September as the trough, fully consistent with Figure 1.

3.2.2 Residual Diagnostics

Figure 3 shows the training residuals over time and their empirical distribution. The residuals are approximately centred at zero with no dramatic heteroscedasticity and a roughly symmetric distribution, indicating that the linear Gaussian assumption is acceptable.

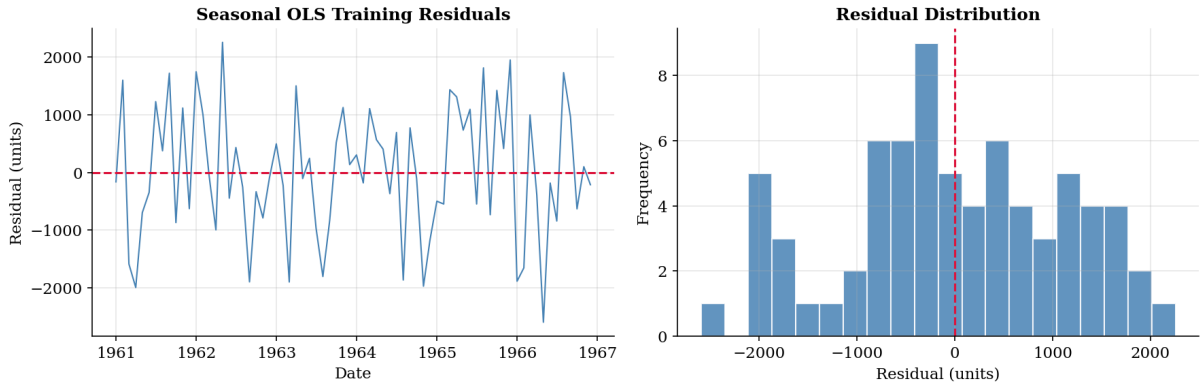


Figure 3: Seasonal OLS training residuals over time in the left and their empirical distribution in the right. No systematic trend or heavy asymmetry is observed; the red dashed line marks zero.

3.2.3 Training and Test Errors

Table 4: Seasonal OLS training and test errors.

Set	MSE	RMSE (units)	MAPE (%)
Training	1.27 B	1 127	7.50
Test	3.82 B	1 954	10.31

The moderate gap between training 1 127 and test RMSE 1 954 units is acceptable for a time-series extrapolation task. Indeed the model generalises well to the 24-month hold-out without severe overfitting.

3.3 Non-linear Models

3.3.1 Random Forest

A Random Forest regressor defined as follow :

- (200 trees, `max_depth` = 6,`random_state` = 42)

is trained on the same feature matrix. Figure 4 displays the resulting feature importances. We thus have that the twelve-month lag y_{t-12} dominates, followed by the time trend t and the one-month lag y_{t-1}.

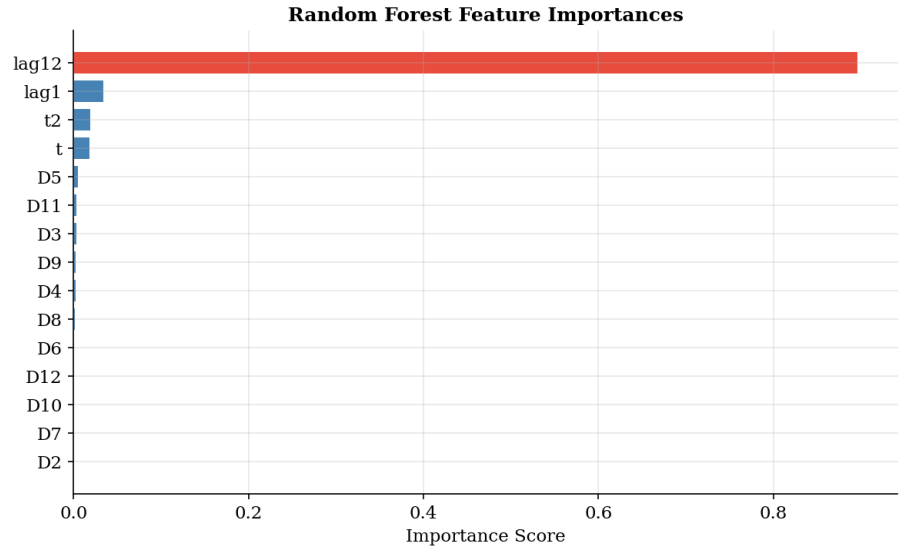


Figure 4: Random Forest feature importances. The twelve-month lag y_{t-12} is the dominant predictor, followed by the time trend t and the one-month lag y_{t-1}. Red bars highlight the most influential features with an importance of $> 8\%$.

While Random Forest attains the lowest test MAPE of about 9.99%, its RMSE of about 2 208 units is slightly higher than Seasonal OLS. Tree-based ensembles add no systematic advantage on this well-structured, low-dimensional series.

3.3.2 Gradient Boosting

A Gradient Boosted regressor of 200 estimators, with learning rate of 0.05, and `max_depth` = 4 achieves a test RMSE of 2 469 units and MAPE of 10.40%. Its performance falls below OLS, likely because the boosted trees over-fit subtle noise in the training set.

3.4 Model Comparison

Table 5: All predictive models — test set performance (bold = best result in each column).

Model	RMSE (units)	MAE (units)	MAPE (%)
MA (3-month)	4 269	3 513	19.25
MA (12-month)	4 009	3 348	19.09
Holt-Winters	2 138	1 788	11.45
Seasonal OLS	**1 954**	**1 665**	10.31
Random Forest	2 208	1 780	**9.99**
Gradient Boosting	2 469	1 919	10.40

The Best model is the Seasonal OLS with the test RMSE = 1 954 units, with the MAPE = 10.31%. Its success stems from combining polynomial trend, deterministic seasonal dummies, and autoregressive lags in a single parsimonious specification. Figures 5 and 6 visualise the RMSE ranking and the test-period forecasts, respectively.

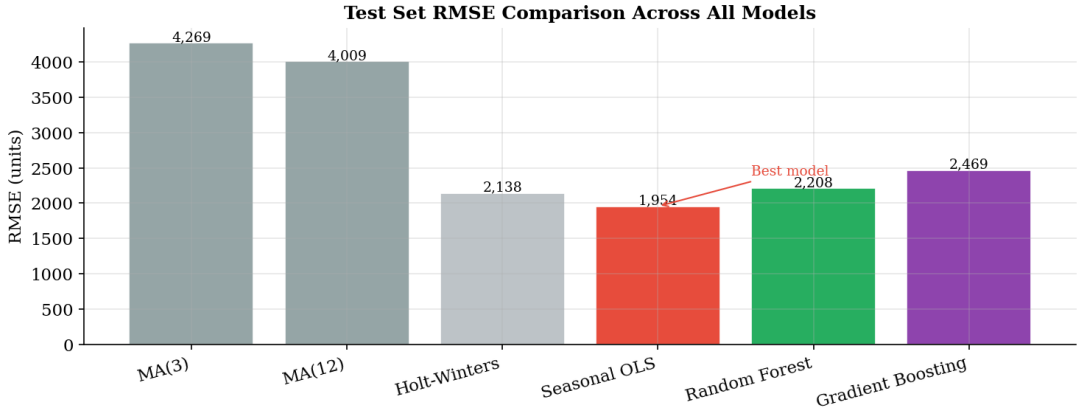


Figure 5: Test-set RMSE comparison across all models. The Seasonal OLS model red bar achieves the lowest RMSE; the arrow indicates the best model.

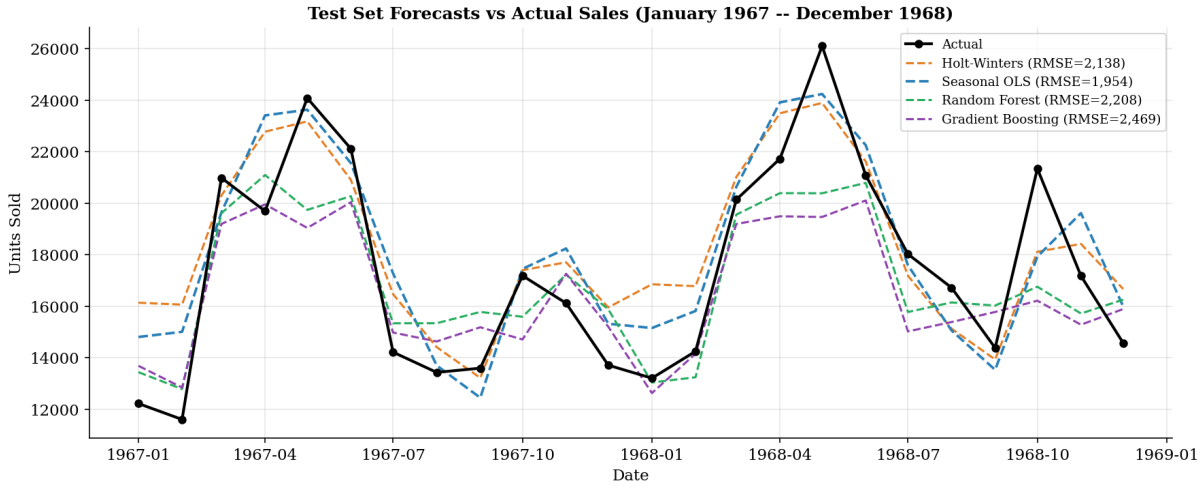


Figure 6: Test-set forecasts vs. actual sales (January 1967 – December 1968). Seasonal OLS (blue dashed) tracks the actual series most closely. Holt-Winters (orange) follows closely. Random Forest and Gradient Boosting capture the seasonal shape but show slightly higher variance.

4 Prescriptive Problem: Newsvendor Inventory Optimisation

4.1 Problem Statement

The prescriptive question is: *how many automobile units should a dealer keep in stock each month to achieve a 90% fill-rate service level at minimum expected cost?* This is a period-by-period stocking decision where unsold inventory is not easily carried forward. We thus have that the newsvendor model is therefore the natural framework.

4.2 Model Formulation

Let $\hat{y}_t$ be the OLS point forecast for month t, and σ the standard deviation of forecast errors estimated as the in-sample training RMSE. We have $\sigma = 1\,127$ units. The objective is:

$$\min_{S_t} \; c_u \, \mathbb{E}\big[(y_t - S_t)^+\big] + c_o \, \mathbb{E}\big[(S_t - y_t)^+\big],$$

where $c_u = \$4\,500$ is the unit underage cost: lost-sale margin and $c_o = \$500$ is the unit overage cost: holding and financing. Assuming normally distributed forecast errors, the critical-ratio result gives:

$$\frac{c_u}{c_u + c_o} = \frac{4\,500}{5\,000} = 0.90 \implies \frac{c_o}{c_u} = \frac{1}{9},$$

and the optimal stock level is:

$$\boxed{S_t^* = \hat{y}_t + z_{0.90}\,\sigma \approx \hat{y}_t + 1.282 \times 1\,127 \approx \hat{y}_t + 1\,444 \text{ units.}}$$

4.3 Results

Table 6: Newsvendor prescriptive results — test horizon (January 1967 – December 1968).

Metric	Value
Forecast standard deviation σ	1 127 units
Safety factor $z_{0.90}$	1.282
Monthly safety buffer $z\,\sigma$	1 444 units
Average optimal stock level S^*	19 537 units/month
Average actual demand	18 093 units/month
Empirical service level achieved	87.5% (21 of 24 months)
Average monthly expected cost	\$1 608 936
Unit underage cost c_u	\$4 500
Unit overage cost c_o	\$500

The newsvendor solution prescribes stocking approximately 1 444 units above the point forecast each month. Over the 24-month test horizon, actual demand falls below S^* in 21 of 24 months, yielding an empirical service level of 87.5% which is close to the 90% target. The small shortfall is attributable to three unusually high-demand months in which actual sales exceeded the upper confidence bound.

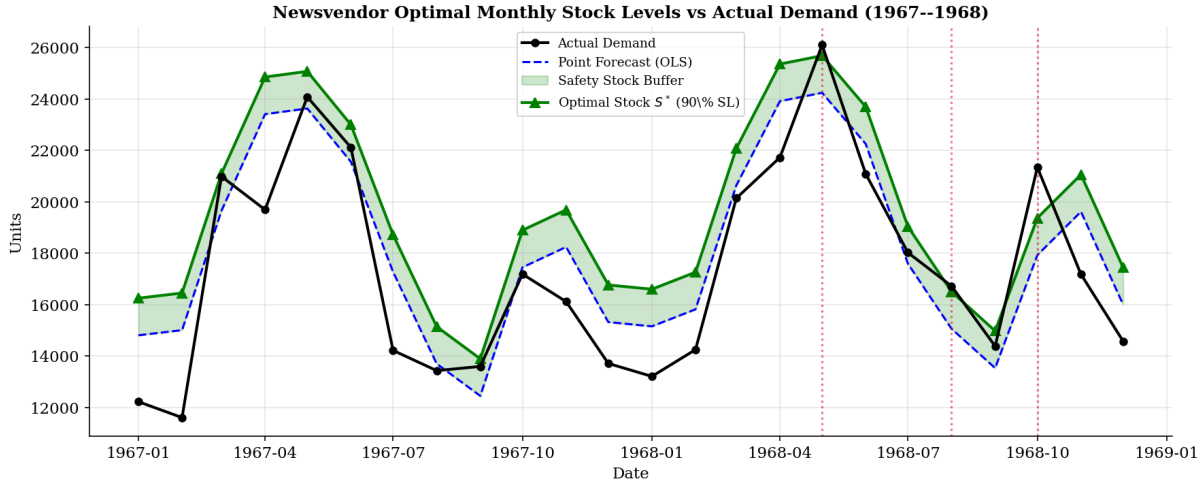


Figure 7: Optimal monthly stock levels S^* (green triangles) vs. actual demand (black) and OLS point forecasts (blue dashed). The shaded green area is the safety stock buffer. Vertical dotted red lines indicate the three months where demand exceeded S^*.

4.4 Sensitivity Analysis

Figure 8 shows how the safety factor z, the optimal stock S^*, and the expected monthly cost respond to changes in the cost ratio c_o/c_u. As overage costs rise relative to underage costs, the critical ratio decreases, the service-level target falls, and the optimal stock drops which correctly reflect the diminishing value of holding excess inventory.

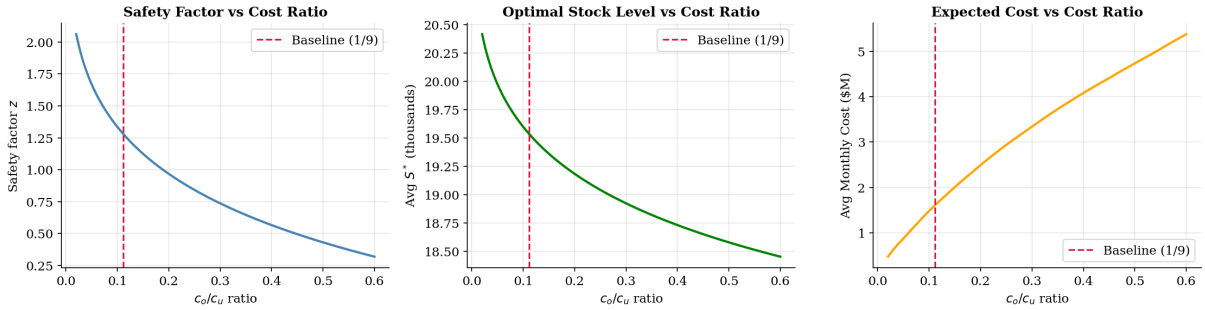


Figure 8: Sensitivity analysis of the newsvendor solution. Left: safety factor z as a function of c_o/c_u. Centre: average optimal stock S^* (thousands of units). Right: average monthly expected cost (millions of dollars). The red dashed line marks the baseline ratio $c_o/c_u = 1/9$.

5 General Assessment

Table 7: Summary of all models — training and test performance.

Model	Train RMSE	Train MAPE	Test RMSE	Test MAPE
MA (3-month)	–	–	4 269	19.25%
MA (12-month)	–	–	4 009	19.09%
Holt-Winters	–	–	2 138	11.45%
Seasonal OLS	1 127	7.50%	**1 954**	10.31%
Random Forest	459	3.02%	2 208	**9.99%**
Gradient Boosting	327	2.14%	2 469	10.40%

5.1 Predictive Stage

Both benchmark approaches of Moving Average and Holt-Winters confirm the strong seasonal structure of the series but are limited by their purely extrapolative nature. Holt-Winters is the best benchmark with the RMSE 2 138, validating the benefit of explicitly modelling trend and seasonality.

The best model is the Seasonal OLS with a test RMSE 1 954 units, MAPE 10.31%. The combination of polynomial trend (t, t^2), eleven monthly dummies, and autoregressive lags (y_{t-1}, y_{t-12}) is a parsimonious yet powerful specification that generalises well to the 24-month hold-out.

The non-linear tree-based models of Random Forest and Gradient Boosting do not surpass OLS in RMSE on this short, well-structured series. The dominant patterns is thus the seasonal cycle and near-linear trend which are efficiently captured by linear terms. Gradient Boosting in particular shows signs of over-fitting to training-set noise. We have :

- train RMSE 327,
- test RMSE 2 469.

5.2 Prescriptive Stage

The newsvendor formulation provides a direct, interpretable bridge between the predictive and prescriptive stages. Stocking $\hat{y}_t + 1.282 \times \sigma$ units per month achieves an empirical service level of 87.5% on the test horizon, close to the 90% target at an average monthly expected cost of \$1.6 M. The sensitivity analysis offers management a clear lever: as overage costs grow relative to underage costs, the recommended stock level decreases in a smooth, monotone fashion.

Three key lessons emerge from this project:

1. At first, Combining deterministic and stochastic components matters. Indeed, seasonal OLS outperforms all purely extrapolative benchmarks because it jointly captures the fixed seasonal shape via dummies and autocorrelated dynamics via AR lags.
2. Secondly, non-linear complexity does not always help. On short, well-structured time series, tree-based ensembles add computational cost without improving test RMSE, illustrating the bias-variance trade-off.
3. Finally we have that prediction uncertainty is operationally valuable. Embedding the training RMSE as σ in the newsvendor formula translates forecast quality directly into inventory decisions, and the sensitivity analysis quantifies how cost structure affects the optimal policy.

References

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